Inclusive Employment Platform

Requirements Engineering Report

Group coursework · SWE2033 Requirements Engineering · Choong Wan Jun, Hooi Ming Han, Tey Tiong Sheng, Yap Jin Ming

1.0 Project Scoping & Ethical Stakeholder Mapping

1.1 Stakeholder Register

Stakeholders	Role and interest	Main needs	Ethical concerns	Engagement method
Job seekers with physical disabilities	Users seeking accessible employment	Workplace-accessibility information and accessible employment opportunities	Mobility barriers, discrimination and unwanted disability disclosure	Interviews, surveys and usability testing
Job seekers with visual or hearing disabilities	Users relying on assistive technologies	Screen-reader compatibility, captions and transcripts or other alternative communication method	Poor interface design could potentially exclude them from the platform	Accessibility testing with real users and assistive technologies
Job seekers with learning or cognitive disabilities	Users experiencing difficulties with complex information	Simple language and clear instructions	Complicated forms could create unequal access or cause users to leave the platform	Easy-reading surveys, supported interviews
Job seekers with mental health conditions	Users experiencing anxiety, stress or mental health problems	Flexible communication	Disability labels could lead to stigma or unfair automated decisions	One-to-one interviews and anonymous surveys
Employers	Advertise vacancies and recruit workers	Job-posting tools and candidate searching	Might intentionally or unintentionally discriminate	Conduct interviews, workshops and prepare surveys
Platform administrators and customer support	Manage accounts and generate reports and ensure platform safety	Moderating tools and accessibility training	May access sensitive information of users or make inconsistent moderation decisions	Internal workshops and operational interviews
Disabilities NGOs or employment support organisations	Support job seekers and advises employers	Consent-based support access and reliable verified information about employers	Must respect user independence and avoid making decisions on behalf of users	Expert interviews and requirements workshops
Government employment agencies	Promote employment inclusion	Transparent reporting, accessibility compliance and accountability	Excessive data sharing could create privacy concerns	Policy review and expert consultation
Technology, cloud and AI	Provide hosting, communications	Clear technical requirements, secure	May introduce bias or excessive data	Technical review

service providers	and security services	data handling and performance standards	collection, causing security risks	

1.2 Preliminary Ethical Risk Radar

Risk	Stakeholders affected	Likelihood	Impact	Risk level	Preliminary mitigation
Unauthorised disclosure of disability information	Job seekers	Medium	Very high	Critical	Optional disclosure, informed consent, role-based access controls, encryption, data minimisation and deletion controls
Algorithmic discrimination	Job seekers	Medium	Very high	Critical	Skills-first matching, regular fairness testing, human review and recommendation explanations
Inaccessible user interface	Users with visual, hearing, mobility or cognitive disabilities	Medium	Very high	Critical	Accessibility standards, assistive-technology testing and participation from users with disabilities
Discriminatory job advertisements	Job seekers	High	High	Critical	Posting guidelines, automated flagging, human moderation and reporting mechanisms
Fake employers or employment scams	All job seekers	Medium	High	High	Employer verification, scam warnings, suspicious-posting detection and user reporting
Pressure to disclose disability	Job seekers	Medium	High	High	Disability fields remain optional and employers cannot require disclosure during initial application
Unauthorised caregiver access	Job seekers and caregivers	Low	High	Medium	Time-limited consent, permission controls, activity logs and immediate access revocation
Excessive or stressful notifications	Users with psychosocial or cognitive disabilities	Medium	Medium	Medium	Customisable frequency, quiet mode and clear notification controls
Complex application forms	Users with cognitive disabilities or low digital literacy	High	Medium	High	Plain language, progress indicators, save-and-return and assisted application support
False inclusive-employer claims	Job seekers and genuine inclusive employers	Medium	High	High	Evidence-based badges, periodic reviews and a mechanism for employees to report inaccurate claims

2.0 Elicitation with Ethical Sensitivity

2.1 Elicitation Evidence Pack

2.1.1 Elicitation Methodology

Technique 1 — Simulated semi-structured stakeholder interview

Purpose:

To obtain qualitative information about accessibility barriers, privacy concerns, accommodation needs, employer requirements and ethical concerns.

Technique 2 — Simulated survey

Purpose:

To obtain structured feedback regarding the importance and prioritisation of accessibility, privacy, job matching, interview and safety features.

Methodology

The two techniques were analysed together. Qualitative statements from the stakeholder interviews were compared with recurring patterns in the survey responses. Findings supported by both techniques were given higher confidence when deriving the preliminary system requirements.

Note on elicitation data: The stakeholder interviews and survey responses in this report were simulated for coursework purposes and were not collected from real participants. The findings demonstrate the elicitation and requirement-derivation process; they do not represent verified views of real stakeholders.

Limitation: The visual/hearing disability, psychosocial/mental health and job coach groups were covered by the survey only, with no interview evidence. Requirements traced only to these groups rest on survey evidence alone. A future iteration would validate all findings through elicitation with real stakeholders.

2.1.2 Stakeholder Participation List

Stakeholder group	Interview (simulated)	Survey (simulated)	Main perspective
Physical disability	✓	✓	Physical/workplace accessibility
Visual/hearing disability	X	✓	Assistive technology/accessibility
Cognitive/learning disability	✓	✓	Simplicity and application process
Psychosocial/mental health	X	✓	Privacy, flexibility and discrimination
Employer/HR	✓	✓	Recruitment and employer requirements
Job coach/support worker	X	✓	Assisted employment and support

2.1.3 Simulated Stakeholder Interview Findings

Based on the simulated interviews conducted, the findings can be summarised as such:

Evidence ID	Stakeholder	Evidence extracts	Interpretation
INT-01	Physical disability	Workplace accessibility information should be available before applying.	The platform must provide accurate workplace-accessibility information, user-controlled disability disclosure.
INT-02	Cognitive disability	Job information should use plain language, and application should be divided into smaller steps. Users are also should be able to save drafts and control notification.	Application process must accommodate different cognitive needs through simplicity, flexible notification and consent-based support.
INT-03	Employer	Assess applicants based on skills and qualification before viewing disability information.	Recruitment itself should prioritise fairness, protect sensitive information and establish trust through employer verification

*Full transcripts of the simulated interviews are available in the elicitation folder of the project repository.

2.1.4 Simulated Survey Findings

Based on the simulated survey responses, the findings can be summarised as such:

Finding ID	Survey finding	Evidence strength
SF-01	Accessibility information is highly valued	High
SF-02	Users require accessible navigation, assistive-technology support and alternative communication formats.	High
SF-03	Users want to save and resume applications	High
SF-04	Disability disclosure should be optional	High
SF-05	Skills should be prioritised over disability	High
SF-06	Users want transparency in job recommendations	Medium–High
SF-07	Flexible interview methods are desirable	High
SF-08	Supporter access requires user consent	High
SF-09	Discrimination, scams and misleading information should be reportable	High
SF-10	Employer verification is important	High

*Full simulated survey response data is available in the elicitation folder of the project repository.

2.1.5 Requirement Derivation Matrix

Req ID	Requirement	Interview Evidence	Survey Evidence	Stakeholder	Ethical consideration
FR-ACC-01	The system shall display workplace accessibility information on job postings.	INT-01	SF-01	Job seekers	Equity

NFR-ACC-01	The system shall support keyboard and screen-reader navigation.	—	SF-02	Users with disabilities	Accessibility
NFR-ACC-02	The system shall provide captions, transcripts or equivalent text alternatives for relevant content.	—	SF-02	Users with visual/hearing disabilities	Accessibility
FR-APP-01	The system shall allow users to save incomplete job applications and resume them later.	INT-02	SF-03	Job seekers	Accessibility
FR-NOT-01	The system shall allow users to customise notification frequency and enable quiet mode.	INT-02	—	Users with cognitive disabilities	Well-being / accessibility
FR-PRIV-01	The system shall not require users to disclose a disability.	INT-01	SF-04	Job seekers	Privacy
FR-MATCH-01	The system shall generate job recommendations based primarily on candidate skills and preferences.	INT-03	SF-05	Job seekers/employers	Fairness
FR-MATCH-02	The system shall explain the main factors influencing job recommendations.	—	SF-06	Job seekers	Transparency
FR-SUP-01	Supporter access shall require user consent.	—	SF-08	Job seekers/supporters	Autonomy
FR-SAFE-01	Users shall be able to report discriminatory or fraudulent job postings.	—	SF-09	Job seekers	Safety
FR-EMP-01	Employers shall undergo verification before posting jobs.	INT-03	SF-10	Employers/job seekers	Trust
FR-EMP-02	Users shall be able to report misleading employer claims.	—	SF-09	Job seekers	Transparency / trust
FR-INT-01	Users shall be able to request alternative interview methods.	—	SF-07	Job seekers	Inclusion
NFR-SEC-01	The system shall protect sensitive personal and disability information.	INT-01, INT-03	—	Job seekers	Privacy / data protection

2.2 Stakeholder Needs Summary

Need ID	Stakeholders need	Primary stakeholder	Source / Evidence	Initial priority
N01	Obtain accurate workplace-accessibility information before applying	Job seekers with physical disabilities	INT-01, SF-01	Must have

N02	Navigate the platform using assistive technologies	Users with disabilities	SF-02	Must have
N03	Access captions, transcripts and text alternatives	Users with visual/hearing disabilities	SF-02	Must have
N04	Use simple applications and save progress for later	Users with cognitive disabilities	INT-02, SF-03	Must have
N05	Control notification frequency and communication preferences	Users with cognitive/psychosocial needs	INT-02, Risk Radar	Should have
N06	Keep disability disclosure optional	Job seekers	INT-01, SF-04	Must have
N07	Receive recommendations based primarily on skills and qualifications	Job seekers	INT-03, SF-05	Must have
N08	Understand why a job was recommended	Job seekers	SF-06	Should have
N09	Choose an appropriate interview method	Job seekers	SF-07	Must have
N10	Control access by supporters or job coaches	Job seekers	SF-08	Should have
N11	Report scams, discrimination and misleading information	Job seekers	SF-09	Must have
N12	Access verified information about employers and inclusivity claims	Job seekers	INT-03, SF-10, Risk Radar	Must have
N13	Keep personal and disability-related information secure	Job seekers	INT-01, INT-03, Risk Radar	Must have

3.0 Requirements Analysis & SD Conflict Identification

3.1 Requirements Catalogue with SD Matrix

Classification: FR = Functional Requirement | NFR = Non-Functional Requirement
Priority: M = Must Have | S = Should Have | C = Could Have
SD flags: S = Social | E = Economic | Env = Environmental

Req ID	Requirement	Type	Priority	SD Flags	SD / Ethical Consideration
FR-ACC-01	The system shall display verified workplace-accessibility information on job postings before application.	FR	M	**S, E**	Improves equal access and reduces unsuitable applications; requires reliable employer-provided information.
NFR-ACC-01	The system shall support keyboard and screen-reader navigation across core platform functions.	NFR	M	**S**	Prevents exclusion of users relying on assistive technologies.
NFR-ACC-02	The system shall provide captions, transcripts or equivalent text alternatives for relevant content.	NFR	M	**S**	Supports equal access for users with hearing, visual or communication needs.

FR-APP-01	The system shall allow users to save incomplete job applications and resume them later.	FR	M	**S**	Reduces cognitive and digital-access barriers and supports flexible application completion.
FR-NOT-01	The system shall allow users to customise notification frequency and enable quiet mode.	FR	S	**S**	Supports user well-being and reduces excessive or stressful communication.
FR-PRIV-01	The system shall not require users to disclose a disability during registration or initial application.	FR	M	**S**	Protects privacy, autonomy and users from pressure or discrimination.
FR-MATCH-01	The system shall generate job recommendation primarily on skills, qualifications and user preferences rather than disability status.	FR	M	**S, E**	Promotes skills-based fairness while maintaining relevant job matching.
FR-MATCH-02	The system shall explain the main factors influencing each job recommendation.	FR	S	**S**	Enables transparency and allows users to identify potentially unfair recommendations.
FR-SUP-01	The system shall require explicit user consent prior to granting account access to supporters or caregivers.	FR	S	**S**	Preserves autonomy and prevents unauthorised access to user information.
FR-SAFE-01	The system shall allow users to submit reports regarding discriminatory or fraudulent job postings.	FR	M	**S**	Provides protection against discrimination, scams and harmful content.
FR-EMP-01	The system shall require employer account verification prior to allowing job posting creation.	FR	M	**S, E**	Builds trust and reduces employment scams while adding verification effort for employers.
FR-EMP-02	The system shall allow users to report misleading accessibility or inclusion claims made by employers.	FR	M	**S**	Supports transparency and protects users from false claims.
FR-INT-01	The system shall allow job seekers to request alternative interview formats during the application process.	FR	M	**S, E**	Promotes inclusive recruitment while requiring employers to accommodate different needs.
NFR-SEC-01	The system shall protect sensitive personal and disability information through appropriate access controls and secure data handling.	NFR	M	**S, Env**	Protects privacy and dignity; minimising unnecessary data also reduces storage and processing.

SD Flag Summary:

The requirements consist mostly of Social (S) considerations as the platform's main sustainability objective is digital inclusion, equality, autonomy, accessibility, as well as protection from discriminatory practices.

Considerations for the Economic (E) aspect are present when accessibility, verification, and inclusive recruitment could affect employer participation or operational effort.

Data handling and processing were the main Environmental (Env) considerations as the elicited evidence establishes data minimisation and reduced unnecessary storage or processing as sustainability concerns.

3.2 Conflict Analysis Matrix

ID	Conflicting Requirements / Interests	SD Tension	Impact	Resolution
C01	FR-MATCH-01 vs FR-PRIV-01	Personalisation vs privacy	Disability data may improve matching but enable discrimination.	Use skills, qualifications and preferences; disability disclosure is not required.
C02	FR-MATCH-01 vs employer recruitment needs	Fairness vs efficiency	Employers may want additional candidate information for screening.	Retain skills-first matching; do not use disability as a primary screening factor.
C03	FR-MATCH-02 vs system simplicity	Transparency vs usability	Detailed explanations may overwhelm some users.	Provide concise, plain-language explanations.
C04	FR-SUP-01 vs assisted employment	Autonomy vs support	Unrestricted supporter access may compromise user autonomy.	Require explicit, revocable consent and limited access.
C05	FR-INT-01 vs employer participation	Inclusion vs economic burden	Alternative interviews may require additional employer resources.	Allow alternative methods where reasonable arrangements are available.
C06	FR-EMP-01 vs employer participation	Trust vs participation cost	Verification may increase onboarding effort.	Apply proportionate verification before job posting.
C07	FR-EMP-02 vs employer reputation	Transparency vs reporting fairness	False reports may unfairly affect employers.	Review reports before corrective action.
C08	FR-SAFE-01 / FR-EMP-02 vs moderation resources	Safety vs economic sustainability	Extensive moderation increases workload and cost.	Combine reporting with automated flagging and targeted human review.
C09	FR-NOT-01 vs timely communication	Well-being vs communication	Reduced notifications may cause missed opportunities.	Allow customisation while retaining essential notifications.
C10	NFR-SEC-01 vs platform functionality	Privacy vs data availability	Excessive data retention increases privacy and storage risks.	Collect necessary data only and apply access/retention controls.
C11	NFR-SEC-01 vs environmental efficiency	Data protection vs environmental impact	Excessive retention increases storage and processing.	Apply data minimisation, retention limits and secure deletion.

4.0 Ethical Framework Application & Fairness Debate

4.1 Chosen Ethical Framework

Deontological Ethics – apply duty-based and rule-driven ethics to system development, ensuring that moral obligations, legal rules, as well as human rights are met by design.

A deontological framework was selected as the platform is directly associated with the rights and dignity of job seekers that are disadvantaged. Evaluation will be based upon whether a

requirement supports privacy, autonomy, equality, dignity, accessibility, and non-discriminatory practices.

4.2 Ethical Evaluation Matrix

Req. ID	Ethical Concern	Evaluation	Decision
FR-ACC-01	Inaccurate accessibility information may disadvantage applicants.	Users have a right to reliable information for informed decisions.	Retain: verify accessibility information.
NFR-ACC-01	Inaccessible navigation excludes users with disabilities.	Equal access is an ethical obligation.	Retain.
NFR-ACC-02	Missing alternatives may exclude users with communication barriers.	Equal access requires accessible information.	Retain.
FR-APP-01	Complex applications may disadvantage some users.	Equal opportunity requires accessible processes.	Retain.
FR-NOT-01	Excessive notifications may affect user well-being.	Users should control non-essential communication.	Retain: add quiet mode/customisation.
FR-PRIV-01	Disability disclosure may enable discrimination.	Users have a right to privacy and autonomy.	Strengthen: disclosure remains optional.
FR-MATCH-01	Disability-based matching may cause discrimination.	Candidates should be assessed fairly.	Strengthen: use skills-first matching.
FR-MATCH-02	Opaque recommendations reduce accountability.	Users deserve transparency in consequential decisions.	Retain: use plain-language explanations.
FR-SUP-01	Unauthorised supporter access undermines autonomy.	User consent must control access.	Strengthen: explicit and revocable consent.
FR-SAFE-01	Harmful postings can expose users to foreseeable harm.	Platform has a duty to provide reasonable protection.	Retain: include review.
FR-EMP-01	Unverified employers increase scam risk.	Users should be protected from avoidable deception.	Retain: proportionate verification.
FR-EMP-02	False employer claims may mislead users.	Users have a right to truthful information.	Retain: review reports before action.
FR-INT-01	Fixed interview methods may exclude qualified users.	Equal opportunity requires reasonable accommodation.	Retain: allow alternative methods.
NFR-SEC-01	Exposure of disability data violates privacy and dignity.	Sensitive information must be protected.	Strengthen: minimise, secure and appropriately delete data.

4.3 Fairness Debate Minutes

4.3.1 Meeting Information

Item	Details
Purpose	Evaluate fairness, justice, and ethical conflicts identified in the conflict analysis and determine suitable requirement resolutions.
Ethical framework	Deontological Ethics
Participants	Choong Wan Jun, Hooi Ming Han, Tey Tiong Sheng, Yap Jin Ming
Date	04 / 08 / 2026
Duration	1 hour 30 minutes

4.3.2 Summarised Debate Record

Issue	Key Perspectives	Agreed Resolution	Req ID
Disability data in matching	Better matching vs. discrimination/privacy risk	Use skills, qualifications and preferences / Disclosing disability remains optional.	FR-MATCH-01, FR-PRIV-01
Recommendation transparency	Accountability vs. cognitive complexity	Provide concise, plain-language explanations.	FR-MATCH-02
Supporter access	Assistance vs. user autonomy	Require explicit, limited and revocable consent.	FR-SUP-01
Alternative interviews	Inclusion vs. employer burden	Allow alternative methods where reasonable arrangements are available.	FR-INT-01
Employer verification	User safety vs. employer onboarding burden	Require proportionate verification before job posting.	FR-EMP-01
Reporting employers	User protection vs. unfair accusations	Reports require review before corrective action.	FR-EMP-02
Accessibility	Inclusion vs. development effort	Keep core accessibility requirements as Must Have.	NFR-ACC-01, NFR-ACC-02,
Data retention	Functionality vs. privacy/storage impact	Apply data minimisation and appropriate retention/deletion.	NFR-SEC-01

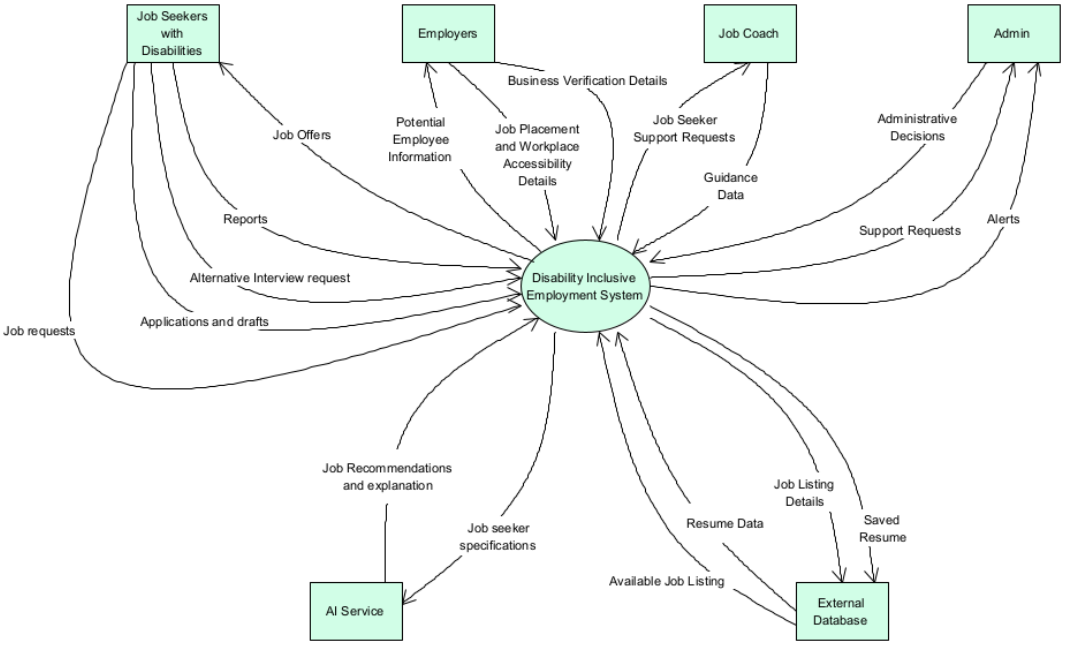
4.4 Revised Requirements List

Req ID	Revised Requirement	Priority	Change / Reason
FR-ACC-01	System shall display verified or employer-confirmed workplace-accessibility information on job postings before application.	M	Strengthened to reduce misleading accessibility claims.

NFR-ACC-01	System shall support keyboard and screen-reader navigation across core platform functions.	M	No change; essential accessibility requirement.
NFR-ACC-02	System shall provide captions, transcripts or equivalent text alternatives for relevant content.	M	No change; essential accessibility requirement.
FR-APP-01	Users shall be able to save incomplete applications, resume them later and continue from their previous progress.	M	Clarifies the save-and-resume function.
FR-NOT-01	System shall allow users to customise notification frequency, enable quiet mode and retain essential service notifications.	S	Balances well-being with receiving important information.
FR-PRIV-01	System shall not require disability disclosure during registration or initial application, and employers shall not require such disclosure at this stage.	M	Strengthened based on privacy, autonomy and disclosure-pressure risks.
FR-MATCH-01	Job recommendations shall primarily use skills, qualifications and user preferences; disability information shall not be used as a primary screening factor.	M	Strengthened to address algorithmic discrimination.
FR-MATCH-02	System shall provide a concise, plain-language explanation of the main factors influencing each job recommendation.	S	Balances transparency with cognitive accessibility.
FR-SUP-01	Supporter or job-coach access shall require explicit user consent, be limited to authorised functions and be revocable by the user.	S	Strengthened to protect autonomy.
FR-SAFE-01	Users shall be able to report discriminatory or fraudulent job postings, with reports subject to appropriate review.	M	Adds review to reduce misuse of reporting.
FR-EMP-01	Employers shall undergo proportionate verification before posting jobs.	M	Balances trust with employer onboarding burden.
FR-EMP-02	Users shall be able to report misleading employer or inclusivity claims, which shall be reviewed before corrective action.	M	Protects both users and legitimate employers.
FR-INT-01	Users shall be able to request alternative interview methods, subject to the employer's available reasonable arrangements.	M	Balances inclusion with operational feasibility.
NFR-SEC-01	System shall protect sensitive personal and disability information through access controls, data minimisation, secure handling and appropriate deletion/retention controls.	M	Strengthened to address privacy and environmental data storage concerns.

5.0 Requirements Modelling & Sustainable Design Principles

5.1 System Context Diagram



5.2 User Stories set with sustainability annotations

User Story:	As a Job Seeker with Disabilities, I want to view workplace accessibility information and requirements on job listings so that I can know if it will be difficult to work there.
Requirement Covered:	FR-ACC-01
Acceptance Criteria:	Given: The Job Seeker has disabilities, When: They view job listing, Then: Workplace accessibility information and job requirements is shown. Given: The Job Seeker is using assistive technology, When: They access the workplace accessibility information, Then: Accessibility information is compatible with supported assistive technologies.
Invest Check:	Independent: - Pass Negotiable: - Pass Valuable: - Pass Estimable: - Pass Small: - Pass Testable: - Pass
Sustainability Annotations:	**Social : Inclusivity, Accessibility**

User Story:	As a Job Seeker with Disabilities, I want to navigate the platform using a keyboard and screen reader, and access text alternatives for multimedia so that I can use the

	platform independently.
Requirement Covered:	NFR-ACC-01, NFR-ACC-02
Acceptance Criteria:	Given: The Job Seeker has disabilities, When: They use only keyboard or screen reader, Then: they can access and operate all interactive elements, navigate between sections, and complete key platform tasks without requiring a mouse. Given: the platform contains multimedia content, When: the Job Seeker accesses that content, Then: equivalent text alternatives, captions, transcripts, or audio descriptions are available where applicable.
Invest Check:	Independent: - Pass Negotiable: - Pass Valuable: - Pass Estimable: - Pass Small: - Pass Testable: - Pass
Sustainability Annotations:	**Social : Inclusivity, Accessibility**

User Story:	As a Job Seeker, I want job recommendations to be driven by my skills with clear explanations of why they were recommended so that I am matched fairly and understand the system's logic.
Requirement Covered:	FR-MATCH-01, FR-MATCH-02
Acceptance Criteria:	Given: the system is generating a list of recommended jobs for the Job Seeker, When: the matching algorithm runs, Then: the algorithm uses skills, qualifications, and preferences, ignoring disability information as a primary screening factor Given: the system displays a recommended job listing, When: the Job Seeker views the recommendation, Then: the system provides a concise, plain-language text string explaining the main factors for the match
Invest Check:	Independent: - Pass Negotiable: - Pass Valuable: - Pass Estimable: - Pass Small: - Pass Testable: - Pass
Sustainability Annotations:	**Social : Inclusivity, Privacy**

User Story:	As a Job Seeker, I want to save applications for job listings so that I can continue customizing them at a later date.
Requirement Covered:	FR-APP-01
Acceptance Criteria:	Given: The Job Seeker is interested in a job listing, When: They want to apply for a

	job, Then: The system allows them to create an application for the job they want. Given: The Job Seeker has an unfinished application, When: They want to stop editing the job application, Then: The system will save the application Given: The Job Seeker has an application saved, When: They open the application, Then: Saved Application can be edited and saved again.
Invest Check:	Independent: - Pass Negotiable: - Pass Valuable: - Pass Estimable: - Pass Small: - Pass Testable: - Pass
Sustainability Annotations:	**Social : Accessibility**

User Story:	As a User, I want to customise my notification frequency and enable quiet mode so that I can manage notifications without missing essential service notifications.
Requirement Covered:	FR-NOT-01
Acceptance Criteria:	Given: The User has notification settings available, When: They customise their notification frequency, Then: The system saves and applies their selected notification frequency. Given: The User has enabled quiet mode, When: the quiet mode period is active, Then: non-essential notifications are suppressed or delayed. Given: Quiet mode is enabled, When: an essential service notification needs to be delivered, Then: the system allows the essential notification to be delivered.
Invest Check:	Independent: - Pass Negotiable: - Pass Valuable: - Pass Estimable: - Pass Small: - Pass Testable: - Pass
Sustainability Annotations:	**Social : Accessibility**

User Story:	As a User, I want Employers to undergo verification before posting jobs so that the job listings that is posted are legitimate.
Requirement covered:	FR-EMP-01
Acceptance Criteria:	Given: An Employer wants to post a job, When: they submit their Employer account for verification, Then: the system performs the required proportionate verification checks. Given: An Employer has passed the required verification, When: they post a job,

	Then: the system allows the job listing to be published. Given: An Employer does not satisfy the required verification criteria, When: they attempt to post a job, Then: the system prevents the job from being published until the required verification is completed.
Invest Check:	Independent: - Pass Negotiable: - Pass Valuable: - Pass Estimable: - Pass Small: - Pass Testable: - Pass
Sustainability Annotations:	**Economic: Verification** **Social : Trust & Safety**

User Story:	As a Job Seeker with accessibility needs, I want to request an alternative interview method so that I can participate in the recruitment process using a suitable arrangement.
Requirement covered:	FR-INT-01
Acceptance Criteria:	Given: The Job Seeker is invited to an interview, When: they require an alternative interview method, Then: the system allows them to submit a request to the Employer. Given: The Job Seeker submits a request for an alternative interview method, When: the Employer receives the request, Then: the Employer can review and respond based on available reasonable arrangements. Given: An alternative interview arrangement is available and agreed upon, When: the Employer confirms the arrangement, Then: the system records and communicates the agreed interview method to the Job Seeker.
Invest Check:	Independent: - Pass Negotiable: - Pass Valuable: - Pass Estimable: - Pass Small: - Pass Testable: - Pass
Sustainability Annotations:	**Social : Inclusivity, Accessibility**

User Story:	As a Job Seeker, I want my registration process to omit mandatory disability disclosures and have my sensitive data protected so that I don’t have to worry about the misuse of my disability and data breaches
Requirement covered:	FR-PRIV-01, NFR-SEC-01
Acceptance Criteria:	Given: The Job Seeker is creating an account or starting an initial application, When: They fill out the required fields, Then: The system does not require or force the user to

	disclose a disability diagnosis. Given: The system collects personal or disability information, When: the information is collected, Then: the system only collects information that is necessary for the intended service. Given: sensitive information is no longer required, When: the applicable retention period has expired, Then: the system securely deletes or disposes of the information according to the defined retention policy.
Invest Check:	Independent: - Pass Negotiable: - Pass Valuable: - Pass Estimable: - Pass Small: - Pass Testable: - Pass
Sustainability Annotations:	**Social : Privacy, Accessibility** **Environmental: Energy Efficiency (less data)**

User Story:	As a Job Seeker, I want to receive help from a supporter or job coach only when I accept so that I can get assistance with finding and applying for suitable jobs.
Requirement covered:	FR-SUP-01
Acceptance Criteria:	Given: The Job Seeker requires assistance with their job search or application, When: They request help from a supporter or job coach, Then: the system allows the Job Seeker to invite or connect with an authorised supporter or job coach. Given: The Job Seeker has connected with a supporter or job coach, When: they request assistance, Then: the supporter or job coach can provide guidance through the functions authorised by the Job Seeker. Given: The Job Seeker no longer wants assistance from a supporter or job coach, When: they remove the supporter or job coach, Then: the system ends the support relationship.
Invest Check:	Independent: - Pass Negotiable: - Pass Valuable: - Pass Estimable: - Pass Small: - Pass Testable: - Pass
Sustainability Annotations:	**Social : Accessibility**

User Story:	As a Job Seeker, I want to report discriminatory, fraudulent, or misleading job and employer information so that inappropriate or inaccurate content can be reviewed and addressed.
Requirement	FR-SAFE-01, FR-EMP-02

covered:	
Acceptance Criteria:	Given: The Job Seeker identifies a job listing or employer information that they believe is discriminatory, fraudulent, misleading, or falsely claims to be inclusive, When: they choose to report it, Then: the system allows them to submit a report and select or describe the reason for reporting. Given: A report has been submitted, When: the system receives the report, Then: the report is recorded and submitted for appropriate review. Given: A report has been submitted, When: an authorised reviewer assesses the reported content, Then: the reviewer can determine whether corrective action is required. Given: the reported content is found to be inappropriate, fraudulent, discriminatory, or misleading, When: the review is completed, Then: appropriate corrective action is taken.
Invest Check:	Independent: - Pass Negotiable: - Pass Valuable: - Pass Estimable: - Pass Small: - Pass Testable: - Pass
Sustainability Annotations:	**Social : Inclusivity** **Economic: Verification**

6.0 Validation Report; Negotiation Summary; Final Requirements Specification

6.1 Validation Report

6.1.1 Validation/Review Techniques

1. Requirements Walkthrough

    - Stakeholders are brought through the requirement list step by step with explanations

2. Validation through prototype

    - Stakeholders are given a prototype built based on the previously elicitated requirements and asked for feedback

Req ID	Requirement	Status	Stakeholder Consulted
FR-ACC-01	The system shall display verified workplace-accessibility information on job postings before application.	Validated	• Physical disability • Employer
NFR-ACC-01	The system shall support keyboard and screen-reader navigation across core platform functions.	Validated	• Physical disability, • Cognitive disability

NFR-ACC-02	The system shall provide captions, transcripts or equivalent text alternatives for relevant content.	Validated	• Physical disability, • Cognitive disability
FR-APP-01	The system shall allow users to save incomplete job applications and resume them later.	Validated	• Physical disability, • Cognitive disability
FR-NOT-01	The system shall allow users to customise notification frequency and enable quiet mode.	Validated	• Cognitive disability
FR-PRIV-01	The system shall not require users to disclose a disability during registration or initial application.	Needs Revision	• Physical disability, • Cognitive disability, • Employer
FR-MATCH-01	The system shall generate job recommendation primarily on skills, qualifications and user preferences rather than disability status.	Validated	• Physical disability, • Cognitive disability, • Employer
FR-MATCH-02	The system shall explain the main factors influencing each job recommendation.	Validated	• Cognitive disability, • Employer
FR-SUP-01	The system shall require explicit user consent prior to granting account access to supporters or caregivers.	Validated	• Cognitive disability,
FR-SAFE-01	The system shall allow users to submit reports regarding discriminatory or fraudulent job postings.	Validated	• Physical disability, • Cognitive disability, • Employer
FR-EMP-01	The system shall require employer account verification prior to allowing job posting creation.	Needs Revision	• Physical disability, • Cognitive disability, • Employer
FR-EMP-02	The system shall allow users to report misleading accessibility or inclusion claims made by employers.	Validated	• Physical disability, • Employer
FR-INT-01	The system shall allow job seekers to request alternative interview formats during the application process.	Needs Revision	• Physical disability, • Cognitive disability, • Employer
NFR-SEC-01	The system shall protect sensitive personal and disability information through appropriate access controls and secure data handling.	Validated	• Physical disability, • Cognitive disability, • Employer

6.2 Negotiation Summary

Requirement	Conflict of Interest	Resolution	Updated Requirement
The system shall not require users to disclose a disability during registration or initial application.	Employers agreed in principle but raised concerns about roles requiring specific physical capabilities for legal or safety reasons.	List physical requirements in the job posting to inform job seekers of the necessary abilities	FR-EMP-03 (new; complements FR-PRIV-01, which is retained): The system shall require employers to clearly list the capabilities required for a job posting before it can be published.
The system shall require employer account verification prior to allowing job posting creation	Strict verification will prevent bad actors from abusing the platform but may deter small-to-medium enterprises (SME) due to requiring heavy auditing	Automatic verification system using corporate email domain and business registry	FR-EMP-01 (revised): The system shall automatically verify employer credibility using email domain and business registry, which will allow employer from that business to create job postings
The system shall allow job seekers to request alternative interview formats during the application process.	Alternative interview formats bypasses skill testing required for job	Alternative interview methods must be granted unless the methods does not benefit the employer in assessing candidate's ability	FR-INT-01 (revised): The system shall require employers to provide multiple interview formats that allows fair assessment of candidates.

6.3 Final Requirements Specification

Part 1: Introduction

Purpose

- To help people with disabilities find work

Scope

- A platform that allows job posting for disability friendly work
- Recommendation algorithm to help provide recommended job posting
- Resume Application module
- Notifications
- Compatible with assistance functions
- Administrative functions

- Supporter module

Stakeholders

- Job seekers with disabilities (physical, mental, cognitive)
- Employers
- Job Coaches / supporters
- Platform Administrator
- AI service providers

Part 2: Functional Requirements

Req ID	Requirement	SD flags	SD / Ethical Consideration
FR-ACC-01	The system shall display verified or employer-confirmed workplace-accessibility information on job postings before application.	**S**	Improves equal access and reduces unsuitable applications; requires reliable employer-provided information.
FR-APP-01	The system shall allow users to save incomplete applications, resume them later and continue from their previous progress.	**S**	Reduces cognitive and digital-access barriers and supports flexible application completion.
FR-NOT-01	The system shall allow users to customise notification frequency, enable ‘quiet mode’ and retain essential service notifications.	**S**	Supports user well-being and reduces excessive or stressful communication.
FR-PRIV-01	The system shall not require disability disclosure during registration or initial application, and employers shall not require such disclosure at this stage.	**S**	Protects privacy and autonomy, and shields users from disclosure pressure or discrimination.
FR-MATCH-01	The system’s job recommendations shall primarily use skills, qualifications and user preferences; disability information shall not be used as a primary screening factor.	**S**	Promotes skills-based fairness while maintaining relevant job matching.
FR-MATCH-02	The system shall provide a concise, plain-language explanation of the main factors influencing each job recommendation.	**S**	Enables transparency and allows users to identify potentially unsuitable recommendations.
FR-SUP-01	The supporter or job-coach access shall require explicit user consent, be limited to authorised functions and be revocable by the user.	**S**	Preserves autonomy and prevents unauthorised access to user information.
FR-SAFE-01	The system shall allow users to report discriminatory or fraudulent job postings, with reports subject to appropriate review.	**S, E**	Provides protection against discrimination, scams and harmful content.
FR-EMP-01	The system shall automatically verify employer credibility using email domain and business registry, which will allow employer from that business to create job postings.	**S, E**	Reduces employment scams by adding autonomous verification effort for employers.
FR-EMP-02	The system shall allow users to report misleading employer or inclusivity claims, which shall be reviewed before corrective action.	**S, E**	Supports transparency and protects users from false claims.
FR-EMP-03	The system shall require employers to clearly list the capabilities required for a job posting	**S**	Improves transparency of job

	before it can be published.		requirements
FR-INT-01	The system shall require employers to provide multiple interview formats that allows fair assessment of candidates.	**S**	Promotes inclusive recruitment for job seekers with different needs.

Part 3: Non-functional Requirements

Req ID	Requirement	SD flags	SD / Ethical Consideration
NFR-ACC-01	The system shall support keyboard and screen-reader navigation across all core platform functions.	**S**	Prevents exclusion of users relying on assistive technologies.
NFR-ACC-02	The system shall provide captions, transcripts or equivalent text alternatives for relevant content.	**S**	Supports equal access for users with hearing, visual or communication needs.
NFR-SEC-01	The system shall protect sensitive personal and disability information through access controls, data minimisation, secure handling and appropriate deletion/retention controls.	**S, Env**	Protects privacy and dignity; minimising unnecessary data also reduces storage and processing.

Part 4: Constraints

- The system enforces strict mandatory data entry for employers.
- The system cannot rely on purely automated takedowns for reported content

Part 5: Interfaces

- Simple design that will not overwhelm users
- Icons that label functions clearly

Part 6: Glossary

SD Flags (S, E, Env): stands for Social, Economic, and Environmental, used to show Sustainability framework.

AI (Artificial Intelligence): In the context of this system, refers specifically to the algorithmic recommendation engine used to match users with job postings (FR-MATCH-01).

7.0 Appendix

7.1 Prototype Link

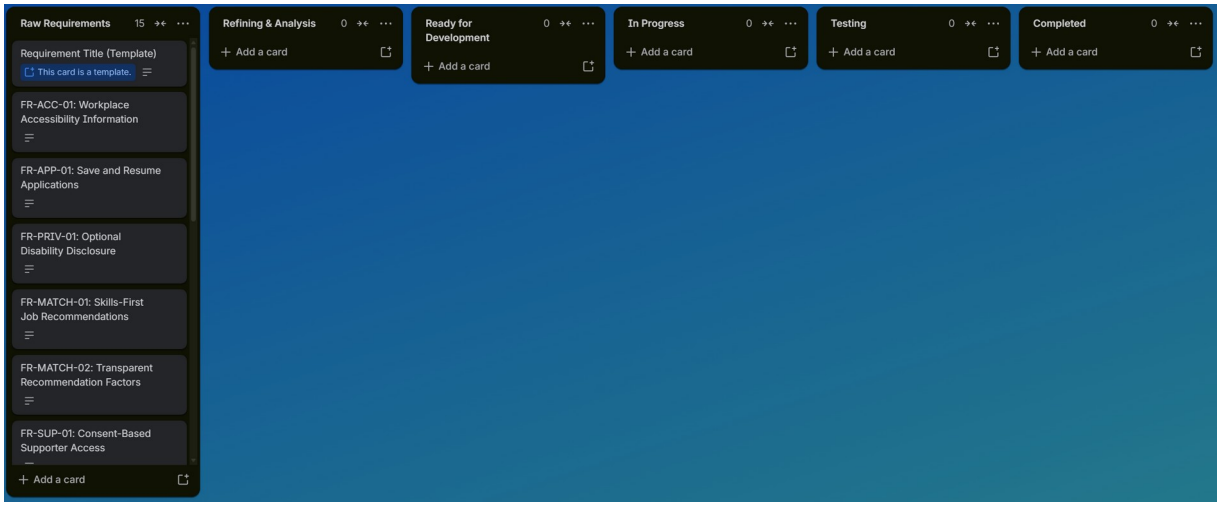
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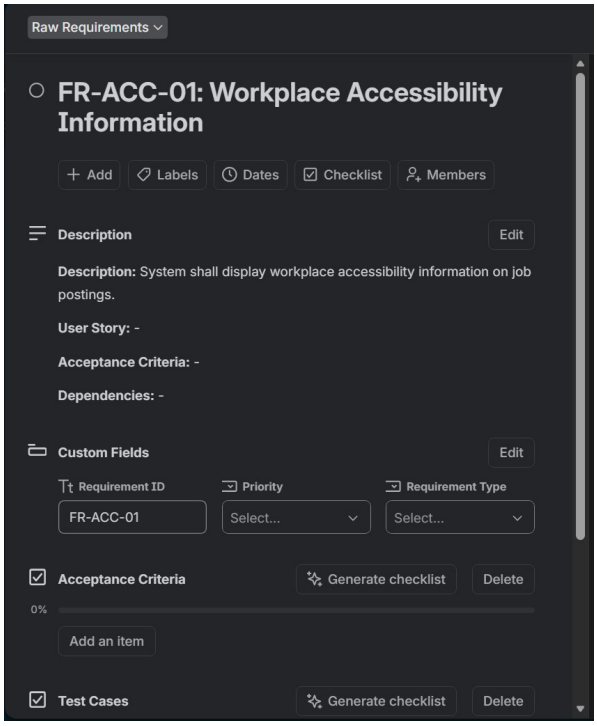
7.2 Trello Board

The team managed tasks on a Trello board. Screenshots of the board at each stage are included in Section 7.3.

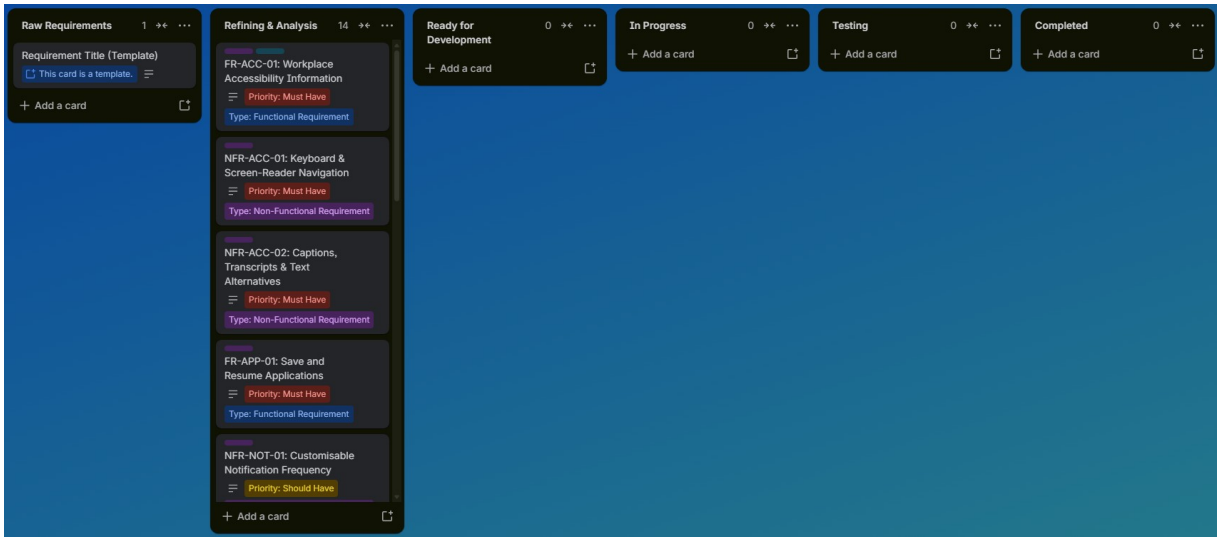
7.3 Trello Board Usage

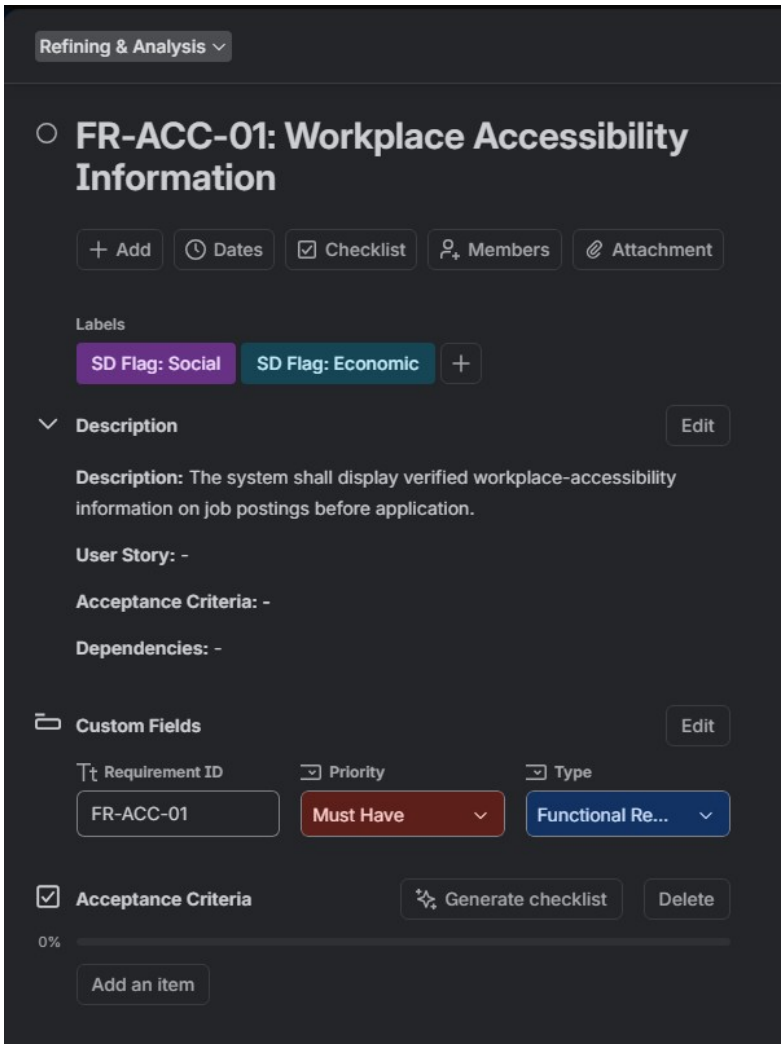
7.3.1 Task 2



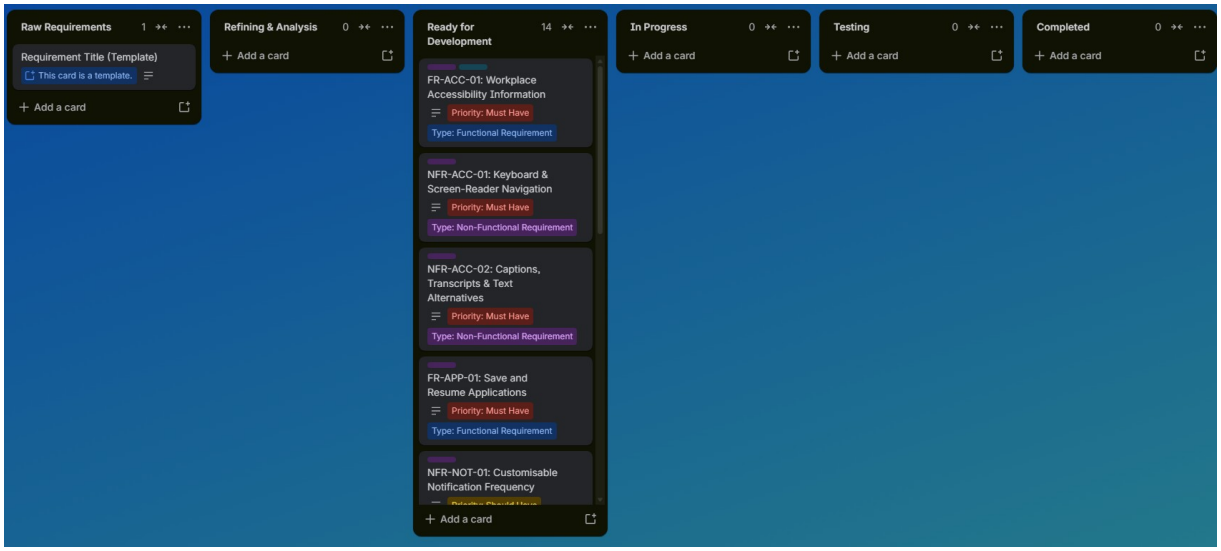


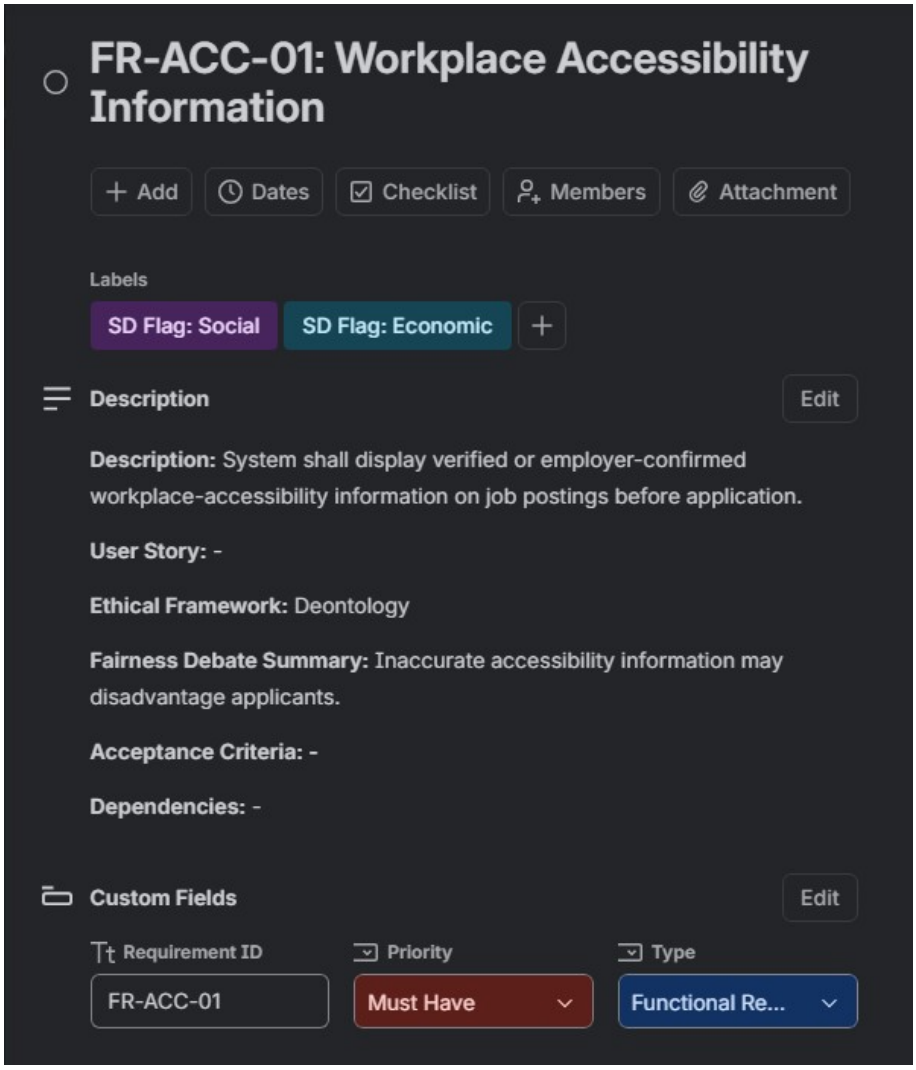
7.3.2 Task 3


Raw Requirements 1
Requirement Title (Template)
This card is a template.
Add a card
Refining & Analysis 14
FR-ACC-01: Workplace Accessibility Information
Priority: Must Have
Type: Functional Requirement
NFR-ACC-01: Keyboard & Screen-Reader Navigation
Priority: Must Have
Type: Non-Functional Requirement
NFR-ACC-02: Captions, Transcripts & Text Alternatives
Priority: Must Have
Type: Non-Functional Requirement
FR-APP-01: Save and Resume Applications
Priority: Must Have
Type: Functional Requirement
NFR-NOT-01: Customisable Notification Frequency
Priority: Should Have
Add a card
Ready for Development 0
Add a card
In Progress 0
Add a card
Testing 0
Add a card
Completed 0
Add a card

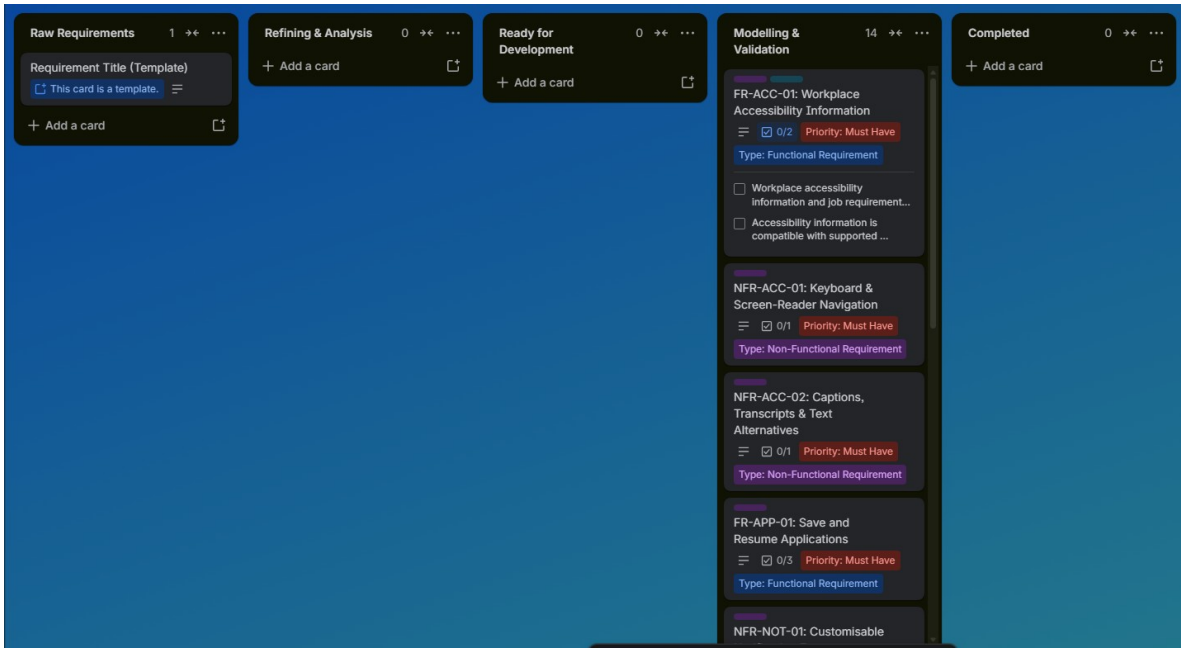

Refining & Analysis
FR-ACC-01: Workplace Accessibility Information
Add
Dates
Checklist
Members
Attachment
Labels
SD Flag: Social
SD Flag: Economic
Description
Edit
Description: The system shall display verified workplace-accessibility information on job postings before application.
User Story: -
Acceptance Criteria: -
Dependencies: -
Custom Fields
Edit
Requirement ID
Priority
Type
FR-ACC-01
Must Have
Functional Re...
Acceptance Criteria
Generate checklist
Delete
0%
Add an item

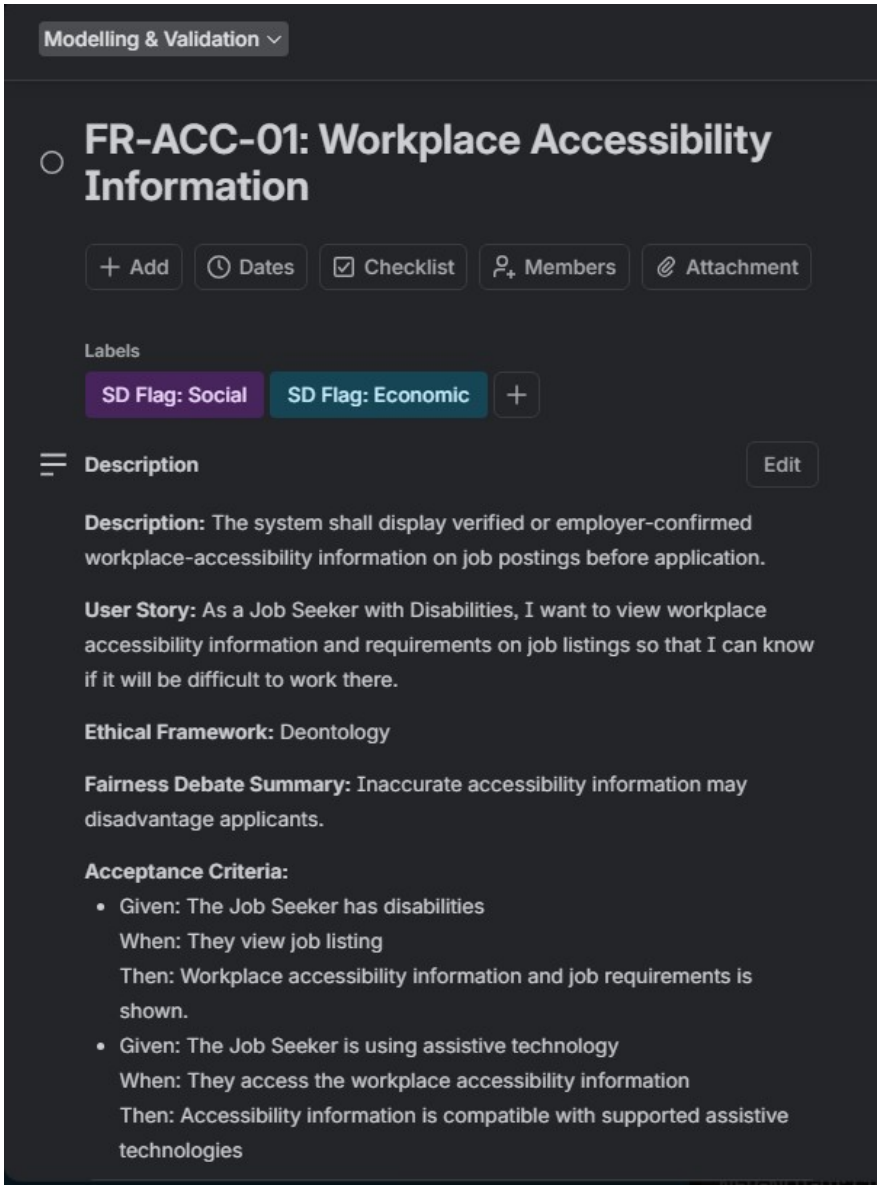
7.3.3 Task 4

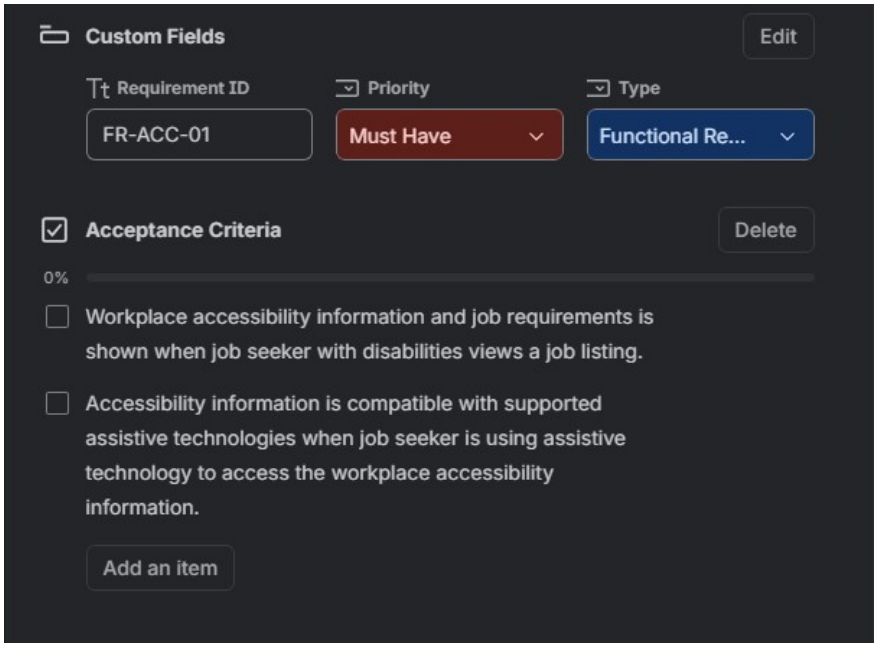

Raw Requirements 1
Requirement Title (Template)
This card is a template.
Add a card
Refining & Analysis 0
Add a card
Ready for Development 14
FR-ACC-01: Workplace Accessibility Information
Priority: Must Have
Type: Functional Requirement
NFR-ACC-01: Keyboard & Screen-Reader Navigation
Priority: Must Have
Type: Non-Functional Requirement
NFR-ACC-02: Captions, Transcripts & Text Alternatives
Priority: Must Have
Type: Non-Functional Requirement
FR-APP-01: Save and Resume Applications
Priority: Must Have
Type: Functional Requirement
NFR-NOT-01: Customisable Notification Frequency
Add a card
In Progress 0
Add a card
Testing 0
Add a card
Completed 0
Add a card


FR-ACC-01: Workplace Accessibility Information
Add
Dates
Checklist
Members
Attachment
Labels
SD Flag: Social
SD Flag: Economic
Description
Edit
Description: System shall display verified or employer-confirmed workplace-accessibility information on job postings before application.
User Story: -
Ethical Framework: Deontology
Fairness Debate Summary: Inaccurate accessibility information may disadvantage applicants.
Acceptance Criteria: -
Dependencies: -
Custom Fields
Edit
Requirement ID
FR-ACC-01
Priority
Must Have
Type
Functional Re...

7.3.4 Task 5 & 6:


Raw Requirements 1
Requirement Title (Template)
This card is a template.
Add a card
Refining & Analysis 0
Add a card
Ready for Development 0
Add a card
Modelling & Validation 14
FR-ACC-01: Workplace Accessibility Information
0/2
Priority: Must Have
Type: Functional Requirement
Workplace accessibility information and job requirement...
Accessibility information is compatible with supported ...
NFR-ACC-01: Keyboard & Screen-Reader Navigation
0/1
Priority: Must Have
Type: Non-Functional Requirement
NFR-ACC-02: Captions, Transcripts & Text Alternatives
0/1
Priority: Must Have
Type: Non-Functional Requirement
FR-APP-01: Save and Resume Applications
0/3
Priority: Must Have
Type: Functional Requirement
NFR-NOT-01: Customisable
Completed 0
Add a card


Modelling & Validation
FR-ACC-01: Workplace Accessibility Information
Add
Dates
Checklist
Members
Attachment
Labels
SD Flag: Social
SD Flag: Economic
Description
Edit
Description: The system shall display verified or employer-confirmed workplace-accessibility information on job postings before application.
User Story: As a Job Seeker with Disabilities, I want to view workplace accessibility information and requirements on job listings so that I can know if it will be difficult to work there.
Ethical Framework: Deontology
Fairness Debate Summary: Inaccurate accessibility information may disadvantage applicants.
Acceptance Criteria:
Given: The Job Seeker has disabilities
When: They view job listing
Then: Workplace accessibility information and job requirements is shown.
Given: The Job Seeker is using assistive technology
When: They access the workplace accessibility information
Then: Accessibility information is compatible with supported assistive technologies



7.4 SuSAF Assessment Table

Req ID	Primary Dimensions	1st Order Effect	2nd Order Effect	3rd Order Effect
FR-ACC-01	**Social**, **Economic**	Renders structured accessibility data on job cards prior to application.	Job seekers avoid applying to physically inaccessible workplaces, saving time and energy.	Encourages employers to upgrade workplace accessibility standards to stay competitive.
NFR-ACC-01	**Social**, **Technical**	UI components adhere to WCAG keyboard focus and ARIA screen-reader standards.	Assistive technology users navigate and apply independently without third-party help.	Reduces systemic digital exclusion for disabled citizens in the job market.
NFR-ACC-02	**Social**, **Individual**	Alternative media formats delivered alongside video/audio content.	Equal access for users with hearing, visual, or sensory processing differences.	Promotes universal design practices as standard across web platforms.
FR-APP-01	**Social**, **Individual**	Application state stored securely in external database for later editing.	Reduces fatigue and cognitive stress for neurodivergent or easily fatigued users.	Higher job completion rates for candidates with chronic illness or limited energy/time.

FR-NOT-01	**Individual**, **Social**	System suppresses/batches non-essential alerts during set quiet windows.	Protects user mental well-being and prevents digital burnout and stress.	Shifts platform design from attention-monetization to user-well-being.
FR-PRIV-01	**Social**, **Individual**	Omits mandatory disability selection fields during onboarding/application.	Candidates retain control over their disclosure and avoid pre-interview hiring bias.	Establishes trust-based hiring pipelines where candidates choose when/if to disclose.
FR-MATCH-01	**Social**, **Economic**	AI engine processes skills/qualifications data while filtering out disability flags.	Ensures skills-first fair candidate matching without subconscious bias.	Breaks down historical employment disparities and systemic underemployment.
FR-MATCH-02	**Social**, **Individual**	System generates plain-language text explaining match score factors.	Enables user transparency, agency, and ability to spot unfair algorithmic bias.	Demands accountability and algorithmic transparency across HR tech solutions.
FR-SUP-01	**Individual**, **Social**	Access control system enforces explicit, revocable consent before linking accounts.	Protects candidate autonomy and privacy while permitting assisted job seeking.	Normalizes supported decision-making models over paternalistic surrogate access.
FR-SAFE-01	**Social**, **Economic**	Submits reported listings to moderation dashboard for administrative review.	Protects vulnerable job seekers from scams, exploitation, and abusive postings.	Cleans up the job market by disincentivizing bad-actor employers.
FR-EMP-01	**Economic**, **Social**	Performs business identity checks prior to granting posting permissions.	Establishes platform safety and trust, though increases onboarding effort for employers.	Reduces fraudulent job scams targeting vulnerable job seekers overall.
FR-EMP-02	**Social**, **Economic**	Submits employer compliance reports for human review and verification.	Holds employers accountable to their advertised workplace accommodations.	Reduces "accessibility-washing" and forces real workplace inclusion improvements.

FR-INT-01	**Social**, **Economic**	Communication tool delivers structured accommodation requests to employers.	Gives job seekers equal footing to perform well in interviews.	Shifts corporate hiring culture toward flexible, inclusive interview standards.
NFR-SEC-01	**Social**, **Environmental**	Encrypts disability data and enforces automated data retention/deletion rules.	Protects sensitive privacy while reducing unnecessary data storage/processing.	Minimizes energy consumption and prevents catastrophic data leak harm.

7.5 Presentation Video

https://drive.google.com/file/d/1uJf7n99MVcz-jms-Q7LPuz4cFs_mEnGU/view?usp=sharing